

ActInf GuestStream 077.1 ~ Gareth Lee Stubbs: "The Police Hunch" Bayesian Brain, Active Inference

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hello and welcome everyone this is active inference guest stream 77.1 we're here with Gareth stubbs going to be discussing the police hunch so Gareth thank you very much for joining looking forward to your presentation and the discussion we'll have thank you so much for having me Daniel and um it's an absolute pleasure to be on the stream one that I've watched many times um and it's privileged to be able to talk through the paper um I'm hoping to give you a little bit of background as to why I wrote the paper where I've come from a little bit of History talk through the paper itself and then hopefully take some interesting questions um if there's anybody watching or um from Daniel himself who's always got some uh very insightful questions to ask um uh so if you're all right Daniel am I okay to start and get going yeah okay so um a little bit of background so I'm a relatively new academic um I was a police officer for 18 years uh in the UK I was in a Northern English Force um and during my career uh I did a wide range of stuff from like Frontline emergency response blue lights to uh community policing which is building relationships with the Community Gathering intelligence and then um uh working on long-term problem solving through to working in uh National strategy uh I worked for the College of policing and um more laterally the home office at the inspectorate and the inspectorate was an organization where we would inspect police officers and make sure that um uh the standards of policing and how they're recorded and monitored and all that was up to a particular level and for the last uh five years um of my policing career I was studying a part-time PhD uh on the side and you would think uh with me having written this paper it would be in a police decision making or similar and it and it wasn't I actually studied um uh using two theories which were uh representative bureaucracy and social embeddedness uh and it was actually about how uh new police recruits uh navigated the Police recruitment process via the use of their social network um it sounds complicated and convoluted and it was um I essentially spent a lot of time uh trying to figure out how people gained advantage or disadvantage through their current social networks whilst trying to get into the police uh with

the Hope um of uh figuring out uh why our um uh recruitment process was so discriminatory um and and I've I've been working on publishing from that PhD uh I've only just finished actually uh over this last maybe two three years uh whereupon I went out to the Middle East to teach police officers now and there's two people that I want to discuss during this stream the first being um uh two different uh groups of people the first one being Professor friston who was just amazing when I contacted him uh and the other being two uh very well-known criminologists called uh Ken peas um and Jason roach and this entire like uh lane or path of study came out of a a discussion with them because I was talking about um very strange decision making that you make while in service so you make decisions that other people look at and say oh that's a very strange decision why did that police officer make that decision but when you speak to the officer they're often able to relate quite a complicated series of past experiences that have led to the decision that um they've made uh and it was K who's again a very well-known criminologist who said to me look you need to read about the Basi and brain he said look this what you're describing is is how the Bas and brain works you know um so he set me on this kind of Down The Rabbit Hole uh Journey if you like um and did a lot of uh reading of the foundational papers around the basing brain which I found very complicated because I didn't have a background in uh mathematics or Neuroscience or cognitive science so it was kind of like jumping in the deep end um but he recommended a really good book called Super forecasters by tlock um some of you may have heard of it some of you may not have heard about it but um it's essentially the human art of prediction like how intuition is made in the basian brain and um I found the book revelatory because uh when I look back at my police experience I could see the development of my basing brain as time went on and how my decision- making changed and this was like kind of mixed because we all have our backgrounds and our history but when I was at the home office in London and we were looking at the way that police forces were organized and governed and all that sort of thing we could see that there was a general tendency towards this kind of promotion of the generalist so a lot of your senior managers seem to have been able to like Bounce Around the organization they've done lots of different things some have been detectives some have been uniformed some have been um uh in corporate some have been Firearms you had all these different like kind of um disciplines if you like um and you ended up with a lot of senior leaders who were jack of all trades and essentially uh masters of non and this was in direct contrast to a lot of the research that looked at you know teams of Specialists people with specialist knowledge that have built up very very solid what you could call like Basi and building blocks when it comes

to specialist decision- making and then when you put them together the matching of all these specialisms then resulted in higher performance of external private companies so we were we were trying to think about how this might look in policing and it was only when I wrote H Read Super forecasting solid blocks of prediction that I started to understand how the brain may work now of course uh if you at any point uh going to YouTube and you type in the basing brain you're going to get professor frisen in some way um so that was my first introduction and this was about two years ago and I absolutely devoured um all of his podcasts I mean everything that I could find find um and that kind of linked in a lot with the llm development obviously we had the side by side active inference seemed to be like picking up steam like crazy and then we had the llms and the AI research picking up steam and then you could see how the two were starting to like twist and merge and there was some really good podcasts from uh machine learning Street talk and and um Kurt J mongal uh podcast Theory of Everything um well we're talking really long form like two and a half three hour discussions on this stuff and I was just absolutely immersed and I started to think about how this has never been used in practice like in policing you know like a lot of the time we go to these training sessions you'd have some experienced practitioners talking to you about how they made decisions we'd have decision- making models and me with my relatively new academic brain was immediately going straight to Google Scholar and saying okay what's the empirical basis for these decision-making models I know everyone's using them but do we know if they work do we know if they speed up our decision making are they better with those who've got better basing building blocks are they not better with so and when you start to look at these questions you realize that there's almost no research um uh to support police decision-making models um uh the best stuff that I found on that is in Andy Clark's book um The Experience machine there a full chapter on looking at police decision making um uh anyway I'm getting slightly distracted how did I actually come to the paper so um there was one experience in my service and I'm going to tell the story because it's in the paper if that's all right Daniel is that okay okay um so I was a newly promoted sergeant and that's like first line supervisor in the police I had a team of about 15 police officers and I was probably about 27 28 years old um and I was out with one of those police officers who was new and he was trying to build his confidence so I said to him look um Jamie I said we're going to go to all the bad stuff tonight so when when the nasty incidents come in we're going to go and I'm going to see if I can work with you on building your confidence in these sort of like you know these these high-risk situations and he was all up for it uh and then a domestic violence incident

came in and the male um offender was still inside the um house and we knew that there had been violent shoes because the person who rang it in was a neighbor and the neighbor had said that she was severely injured and bleeding so uh because there was a a chance that it could get rough and we might need to um make an arrest straight away I thought this would be good for Jamie to go to uh and he was a taser officer and I don't know how how like uh experien you guys are so I'll just let you know but us police officers in England we don't carry guns like um there's probably about five or six% of police officers that carry guns at the most and a lot of them are only sent to incidents that may involve Firearms so the vast majority of incidents we go to we have a Baton which is like a stick um and a uh a spray uh and there're the they're the two bits that we've got to deal with it now Jamie had just got his taser um and uh he was trying to build his confidence um so we turned up there was a very injured lady on the sidewalk she was pulling clumps out of her hair she'd been clearly punched a great deal her face was an absolute mess and she told us that he was still inside the male offender uh and that obviously he will be violent towards us as police officers now that yeah if you read the paper this may come across as if it was a like a kind of um a one-off incident it wasn't you go to things like this all the time on a busy Friday or Saturday night I could go to the two or three of these maybe um so this was very very normal nothing out of the ordinary and I said to Jamie okay well let's go on um we'll wait for some back up because he might he might be violent this guy and you know we'll go in and we'll we'll see what we can do he's obviously going to need to be arrested because of the damage that he' done to the lady um and uh she said no you I've got my one-year-old upstairs and he's in bed and the minute that she said that like we can't wait for backup we have to go in because there's a chance that uh the the child could be injured so obviously that changed our decision- making and as we're working up the front door up to the front door the door opens and this big muscly guy comes to the front door and the thing that got me at that time and again I just have to stress like I got these were very very normal incidents like would go to these all the time there was something about this incident where I felt like my heart hit my stomach big adrenaline dump all of the um uh prickles on my arm stood up like my hair stood up power erection and up the back of my neck and I remember looking at Jamie Jamie didn't feel this and I spoke to him afterwards it was really strange because there was two of was presented with the same incoming sensory stimuli I had a very different bodily reaction um and we talked about it afterwards I mean at the time out like I say I was eight or nine years of experience Jamie was like coming up to two so there was a big buildup of basing priers in in me for some reason um anyway uh let's qu long story short uh

we ended up fighting with the male uh he was very violent towards us really really bad incident um and when we finally got him in handcuffs as we got to the front door we noticed that he dropped a knife and he'd had a knife when I'd pulled him out of the doorway now um this is like one of those moments that you have in the policing career where you're like your mortality punches you in the face you know because had we gone inside or had we engaged him in a very different way there's a chance that both of us would not have come away from that incident and the sometimes that like hits you and it hit us both then and and Jamie got very upset remember it very distinctly we we walked into the kitchen we had to go and check that the young lad was okay and he was and obviously we got the injured party back in to the house uh but I remember doing like an immediate reflection afterwards and saying to Jamie like I have never had this severity of body bodily reaction like that that came on so quickly and I've used the word salans to describe it in the paper but there was something that was distinctly different and I don't know what it was but obviously whatever signals I was receiving be it from the lady on the curb or from the M's Behavior as we approached the door there was something in me that told me interceptive so it was an internal feeling that then triggered my ctive processes to approach the incident differently anyway that incident obviously he went to court he was he went to jail actually and so the incident itself we managed to safeguard the lady in the sun which is always a good outcome um but for me it really stuck with me uh for a couple of reasons the first was the the bodily reaction that was so different was something that Jamie didn't receive at all so there was no difference in him he he saw nothing um of the evidence that I saw and when I say saw this is obviously um and again cover it in the paper there's so much information that you're taking in these incidents and it's in such a temporal window that such a tight temporal window that your ability to be able to affect the environment that you're in is uh very very highly constrained because you're only going to be at that incident maybe for minutes if you're lucky an hour or over depending on how busy the radio is so you can't do what you would call any longitudinal Niche Niche construction your ability to mold or shape your environment when you turn up as a police officer is limited solely to the time that you're there and because you don't know any of the people there there's something going on in the police officers brain and again this is discussed in the paper that is very different to a normal work environment so this is I'm just going to I'll try my best to give you an example so if you work on a till in customer services you have a very predictable way of living your your day out and of course the unpredictability comes from the customers but there's often a set of like procedures or protocols that you follow that enable you to deal

with those people in a particular way and allows you to predict your work and hopefully comfortably do that now when you're a police officer you have the law and this was interesting actually Daniel because when we first had a chat about this I think you put me in touch with somebody who was talking about the law as an agent of active inference which is proper interesting stuff and if we get time like later on it might be worth a discussion but the officers are working within the law so you've got uh bounds if you like you have boundaries that you work within and um there's particular actions you can take but everything is so highly unpredictable I'm not working with a till I'm working directly with people it's often in a very very emotionally charged environment because that's why police get called in the first place you don't know any of the people that you're dealing with and they're not coming to you asking for the same service so it's not like you're going up to a shop and saying I want to buy this again and again and again which is the same thing that you'd get as a cashier when we turn up a lot of the time people don't even want police there so you come in as like a kind of alien if you like and there's often something that again it's like high emotions everything's going a little bit crazy decision making of those people is normally totally out of the norm you're trying to bring everything back within the norm so if you like that the officer is like an agent of allostasis we're trying to bring back homeostasis to the environment when you get there but it's so messy and so difficult that you often end up making uh decisions other people may look at and say why have they done that that's such a strange decision but if you've been and and I suppose in the paper and I go through it if you've been in that particular situation a number of times before and you have a good set of basing building blocks that allow you to make improved decision-making as a result of that experience you may end up with improved outcomes and I'm not going to say that you will but you may and it's a solid Theory to um uh back up good experience Le decision making and there's lots of things that you can look at and hopefully the paper towards the end might be able to just talk about some of those and I can take the viewers through them so that's the kind of background the salience of this particular incident um is the one that the paper is based on it was so different to me during my police career that I needed something that explained it this Theory explains it in a lovely way it's so nice um and and obviously the the kind of absorption of uh Carl Friston's um uh work over time and the reading of the books I mean that's inference book by Thomas P I know that you're doing a series on it a reading group on it it's brilliant I would say it's a little bit hard going don't pick it up as your first book do some listening to the podcast first try and get your head around how it all works and then pick up the book and go through it it gets a little bit heavy

towards the end the experience machines a really good introduction that's another good one Andy Clark talks about active book it's not an active inference book it gives you a much wider view on the uh the cognitive neuroscience and how the active inference stuff is brought in uh but I really enjoyed that uh and then obviously uh when I'd had this idea and I'd written most of the paper I think I was at maybe you know six and a half 7 thousand words I took the leap and I emailed uh Mr priston uh and I and I I just emailed Professor friston with a massive a massive chunk of Humble Pie and said um Professor lovely to meet you I feel like I know you because I've watched you a million times uh online um this is what I've written what do you think and then uh to my surprise um he sent me back uh another 800 words or so um with a very heavy endorsement uh and I couldn't I was a little bit in shock it was one of those moments I was looking at my computer and I was is that is that Emil from oh my goodness and then I opened the manuscript and I was like wow okay this is amazing um and then obviously a little bit more dialogue uh and we chose to uh submit but needless to say um and I mentioned this earlier in the podcast because I don't have a background in maths or Neuroscience um I'm fly sure that those holes were evident in that first draft um and Mr friston did an absolutely amazing job of of plugging them so I guess that's the story uh behind the research Daniel did that did that make sense yes you good with that yeah thank you okay uh have you got any questions so far or do you want me to just work through the paper quickly yeah let's look through the paper okay cool right I'll just go to share screen then um okay okay is that Yep looks good there we okay perfect um so I don't want to uh read the paper only because I think everybody will die of boredom but I will skip through uh the sections and explain to you my thinking um so I I go very gently uh into setting the scene uh talk a little bit about uh how how we police officers are often flung into what you would call Vital High State situations and this is elaborated on a little bit further but it's that temporal nature of the decision- making which I think causes very specific grooves in the police officer's brain and I'm using the word Groove because I think it's a really good analogy that kind of like explains the brain it's not that we can't step out of the groove but they help us make decisions and I suppose that's in line with her istics it's actually in line with like hemispheric Theory like all sorts of like um and I talk about this later like system one system two thinking this is all quite um well established good psychological research it all like kind of melds together it gives you a good little background uh so I talk about a little bit of the history um with regards to uh the hunch and what the hunch is um it's been described using a number of different terms uh so we've got suspicion in here uh This was done on Research that was um conducted when it

comes to things like stop and search and stop and search is such a uh a contentious area of policing so uh um high impact when it comes to things like Community tension Community confidence trust safety legitimacy a lot of it is linked with uh the use or the abuse of stop and search so there's been um quite a lot of study that's looked at the decision to stop and the decision to search how those decisions are created and often through um ethnographies where academics have gone out with police officers and watched um uh officers conduct their um their stops and their searches and how they've uh made they've then asked after that's been done how did you make that decision how have you come to that decision to to stop and search that person what was going on inside your brain and I've then introduced the domestic violence incident which I've already discussed so I'm not going to go through that um but I've just talked about this bit here where it says the scenario creates a fundamental question how does an officer's body generate these hunches these internal alarm bells and what role do they play in navigating the experience of being police officer um now they're not restricted to police officers and I talk about this here especially when it comes to things like um uh medicine uh and Public Service uh based uh professions uh Armed Forces firefighting Etc uh there's uh an excellent article and if I if I could remember it I would share it um but it was about a group of firefighters that went in they're in the midst of like you know uh fighting this crazy big fire and all of a sudden the supervisor just says right get out get out get out and he he grabs all his team and he pulls them out and literally within seconds the floor collapses now obviously there's going to be no uh uh super particular s Salient detail that's made him do that it's going to be a gathering of uh details which are going on often unconsciously in the midst of the fire even him I imagine can't process um in in a way that's rational and deterministic he can't say this led to this and then this led to this and then this led to this it just this overwhelming feeling came to him as a firefighter and he said no come out come out quick and as they came out the floor collapsed now that's an example of the hunch but the hunch within a firefighting circumstance it's the same thing it's like a kind of overpowering feeling of intuition uh so I talk a little bit about this here uh and we talk about it being a a little bit wider than um stopping a vehicle um there is a discussion here of uh um almost like the spectrum of what a hunch can be like how does intuition present itself there is some research coming very shortly from somebody called Emily Quinn at Cambridge um she's just done a PhD in something called The Copper nose which again is the police hunch and I didn't know about this um until after i' written it um but I know that she's done a data collection I know that she's much more hot on the terminology and the neuroscience and the cognitive science behind this

and I know that her research is about to land so uh just for the listeners it's definitely coming that's going to be a treat uh when it lands uh but a hunch or a copper knows when you're trying to find vehicles that may be up to no good for instance uh I'd been in cars with people who are very experienced police officers and they say something like I want to stop that car I don't like the look of it now I'm trying to learn and my immediate question would be like well what do you mean you don't like the look of it what is it about that car that's promoting this feeling of unease and even the most experienced police officers they can't answer that question uh they could they'll say something like oh well maybe it's the manner of driving you know but if it was the manner of driving and I might have picked up on it there's something else there something that's that's uh triggering and I imagine their hunches and their heuristics that have built up over time again it'll be basian uh there will be tiny signs tiny um inputs sensory inputs about a particular vehicle and the way that it's behaving that over time have led to the building up to of this this intuition now that is a very very small slice of what is intuition a very very big slice of what is intuition might come in the detective world so if you're working on a murder case it could be that you have a very particular feeling about one of the suspects and that feeling may build up over time now again very very difficult to explain why do you why do you have a feeling that that suspect did or did not do it why why is it coming and again those detectives and I know I've worked with them they can't explain it there there's something that's that's building up for them and they can feel the hunch the hunch isn't something that's visceral it doesn't come into the pyro erection kind of like you know like response to threat or response to stimulus that's severe it's very very slow creeping and it leads to like this it's almost like intrusive thoughts you know like that they're very very slow they come in without they're unbidden there's no like kind of I'm not very very I'm not happy with this you know this is not good you know it's Uninvited uh so the hunch can manifest itself in lots of different ways from the kind of I want to stop that car I don't like the look of it to something that generates over weeks or months and again is very difficult to discern like where does that sense come from we don't know we don't know and that's one of the things that I think that should be looked at and I discussed at the end when we talk about where the research could go uh so again I take them through um the incident which I've just discussed uh and we talk a little bit about the background of intuition research uh and then we talk about uh rational decision-making intuition there's a meta study there from Wang Wang et al in 2017 he's talking about constructs um I think they are separate they run upon the Dual cognitive system system one system two again emotional thinking what you would call emotional thinking and rational

thinking if you were being a Philistine like me um so yes um and then I I've very very tentatively and I have specifically use that word I'll just highlight it there tentatively with this prior discussion in mind uh the author uh posits a definition of a hunch with the uh for the following discussion which is an often unbidden Salient intuition experienced by a police officer that subsequently influences their actions now when I say influences their actions I don't mean that it it specifically sends them in a particular direction that direction can be anywhere but it's the fact that it causes some sort of change in direction or causes you to consider a change in direction that then uh uh so in other words it's instrumental it's not something that just pops up and it can be dismissed it does something now there was a some uh push back a little bit um on the definition so words were added as we uh went in and through from the reviewers who were amazing uh on the article they were so cool gave lots of different um uh additions or possible additions and talked about this and we ended up with uh this this inclusion of the word salens to kind of gener to describe a hunch you know like what well it's the salience of the hunch it's not like a normal hypothesis our brain is constantly generating hypothesis all the time but there's something about this particular hypothesis when it pops up Wham it hits you okay well what does that hit feel like where where do it come from and why is it different to your normal train of thought what is it that's that's made it pop out and that then leads um uh to the maths now uh some of the narrative probably halfish of this is mine some of it calls uh and and Professor friston went into a lot more detail than me um the uh equation that I put forward for surprise was tripled in length um so he went into a lot more um this was the basing brain bit we introduced the basing brain um the if you like the older research um and then we bring in the newer research active inference and free energy principle uh and then uh Professor friston uh added the detail in the uh equations here and also the figures um which have come in which are the the background if you like for these equations how do these equations work and what do they say uh but uh uh what's the gist so if I was to describe this uh surprisal bounds or the level of surprise that somebody operates within a police officer is going to be very very sensitive uh to boundaries of surprise this is the hypothesis essentially because and I talk about this in the next section which is about uh cultural Niche construction because they have very little control over their environments and they have very little knowledge of the people within them um I I think that their brain will generate hunches in a much more sensitive fashion within those constraints okay so essentially um uh through the Gathering of experience uh very uh temporally informed if you like um work environment if I go to a domestic and I walk into a domestic incident and the male and

female are shouting and throwing things at each other I have I have no discretion I have to take action immediately because uh there's there's harm uh that that can take place at any point to either of the parties okay so I have to uh take what is a very active role in that environment immediately and therefore I'm constantly training my brain I think personally as a police officer this is what I think goes on to take action and to take action decisively in order to reach desired outcomes now in most people's daily business they have lots of time to make these decisions and as a police officer you just don't and that that generates a a uh grooves I think it generates grooves in the police officer's brain and it leads to something which has come out in the ethnographic literature around policing called pragmatism now it's described as a tendency to do over a tendency to think so a police officer will often tend I think towards jumping into some sort a very active role in the environment sometimes too quickly way too quickly you know but I think that that too quickly and I'm going to use air quotes there is basian I think that you your your brain is trained to intervene quickly in many many cases and it sets up grooves and those grooves are very easy to fall into and that presents risk you know so um I'll give you an example if I'm going to be controversial here but this is where I think the research will be really cool if for instance um there is a drug problem uh within a particular neighborhood and that neighborhood is high in ethnic minority and it could be ethnic minority from anywhere okay so um it could be uh uh so in the UK Albanian for instance okay or it could be black in London or it could be afro Caribbean there's lots of different ethnicities but um if there is a community which is heavy in a particular ethnicity and as a police officer you are constantly interacting in what will be um highly temporal Niche situations normally violence related uh with that Community that's going to influence the way that that officer thinks and reacts in other um maybe less temporally influenced um situations so uh I'll give you a physical example if I am going to um uh Gang Related violence a lot okay and I'm dealing with victims that have been injured as a result of Gang Related violence if I do a stop and search in that same area and I have been to all these previous incidents there's going to be a building up of influence and experience in me which is then going to change the way that I behave when I stop that car so and you will see this in the research this comes out in a lot of the um uh the way that police officers are perceived when they're dealing with ethnic minorities the way that police offic officers act they're not the same okay um when it comes to uh majority population interactions okay but I do think and and again I'm pushing this Theory way out from Beyond my paper here I do think that that um there will we can probably find some evidence of why that may be the case uh and and maybe prove empirically that the more

incidence that you go to that could be highly temporally influenced that may involve particular ethnic minorities could then change the way that you as a police officer operate on a much more wider scale with members of that ethnicity now I don't think a lot of that's conscious and I know because I've worked with police officers I mean I met one racist police officer in my entire service of 18 years so um I know people say are the police officers racist you know I don't think it's as simple as that I think it's way more complicated I think that we have to really understand why police officers make the decisions they do over time for us to properly unpick this problem I don't think it's something that we can turn the page on uh and it's it's going to be years of good research some of which is alluded to in Andy Clark's book um so anyway I'm getting a little bit wide here but all of this uh influences that conversation that I've just had I'm not gonna go into the detail but that's that's that's the bit that that is there um and we have this wonderful addition from him uh which gives you the background to the theory uh from Professor Friston uh now this this is the section that if I had the choice I would have put a lot more um words into uh and this is the hunches and nice construction stuff and you'll probably be able to tell that from how I've been talking about it today um but low surprise longitudinal relationships probably make up the vast amount of our uh interpersonal relationships that we all experience on a day-to-day basis all of us um in general in the world unless you're in somewhere that's very very volatile mostly they're going to be low surprise longitudinal relationships so you're going to have more opportunity to mold the environment over time okay uh so that's the and that's the active bit of the inference okay so you've got this like kind of ability to interact with the environment the environment then interacts with you and you end up with this like kind of constantly like forward and back in between you and the environment the problem is of course uh again with police officers the ability to longitudinally uh change or influence an environment is just not there you have I have an hour at most at the incident and then I'm getting sent to another and that may then cause I think this pragmatic lean into to do rather than think so that's the kind of like the the the what I'm getting to in this last this last part here before moving into the discussion so um there's a couple of questions that I've come up here I've mentioned the pragmatism here from Loftus this is from the cultural literature so um what cognitive signals would uh were generated by IAL attendance that contributed to the hunch generation when the basian brain processes these signals what is produced empirically during the process of active inference what was the officer's qualia so how does the hunch present into Consciousness massively complicated and I've even put it in the paper we don't want to get into the the Consciousness debate because it's absolutely

fascinating but it's just incredibly complicated so there are some things here that make that person aware and this refers to that salience which is in the definition how were those qualers interpreted so when the hairs on my arm stood up how did I see the hairs on my arm standing up so obviously you're going to receive the signal but how do you interpret and then turn that signal into action these are questions that haven't really been answered yet within this particular area and then um this last one which I think I mean it looks like an afterthought because it's one sentence but it's not one sentence it's actually really important how did the officer justify this hunching latest Court proceedings and I have written hundreds probably thousands of police statements over the years and I know that reading back those statements they will read very rationally they would read super rationally so my ability to rationalize my decision- making post decision making is way better than my ability to interpret a hunch rationally and there's a big disconnect there so um uh you end up with a lot of obviously hindsight virus if you stopped a car and you found some heroin well you stop the car because you think that there was drugs on the car but as mentioned earlier in the um article sometimes that hunch is non-specific it's raised through experience that's that's very very diff different really different and I don't think the officers have got the skills to interpret the hunches well enough to make well reasoned decisions I think that they rely upon um intuition much more than they say they do and I know well I know because I've been one and I've been around them I know how these things work so I just don't think that it's been researched well um and then as we get towards the end um uh we move into the again the the theory uh support from that figure and we posit a series of areas for future research so we've got here long Udin study on hunch development so this is tracking the development of hunches in the in police offices as their career develops I think this is super interesting because when I talked about going up to that door in the domestic and Jamie not feeling anything and me having the most Salient feeling I've ever had there's a there's something going on there you know like so when does that start to happen like when do when do Police intitions become so noticeable and Powerful that people can see them and be able to act upon them will be able to notice them and suppress them because even that is an interesting area so if you are getting a hunch and you think that hunch may be informed through past discriminations then surely you need to be able to suppress it you know you shouldn't be acting upon it um so if you think for instance that um uh there's an element of discrimination in stopping a vehicle you need to be able to notice that and stop that before it happens you know and it could be unbidden and again this is in the definition so you haven't invited the hunch it just comes but you may know that there's not

enough there for you to make that stop there's not enough evidence so something's going on and you need to be able to see that and stop happening uh comparative analysis with other professions so this is looking at the generation of hunches for instance in the medical environment neurobiological correlat so this was to look at um neurobiological processes associated with hunch formation uh the use of brain and body Imaging techniques this is probably way above my pay grade but it would be super useful and there is some precedent for it in very very primitive studies in the 70s and 80s that look at things like um heart rate and blood pressure over the Tour of Duty of a police officer you know um but it'd be brilliant to see that Advance much more advanced wearables even looking at good scanning equipment how are these countries generated it might even be one for the future but we thought we'd discuss it um the impact of training on hunch accuracy this is something that came from Professor kpes he was talking about training scars um have you heard of training scars Daniel no so training scars are where you constantly train in a particular way your grooves become very very pronounced in the way that you approach similar incidents because you train so much and then when an incident presents itself your brain kicks you into those grooves even though the grooves may be inappropriate so they can lead to um uh for instance taking a shot because that's a shot you would have taken in training despite the fact that that shot was non-justified or another one where there's not enough threat in the training scenarios so because of that you act in a much more uh a slow and considered way when you need to act much more quickly both of those will be considered training scars and again um uh so overtraining may suppress good hunch Intuition or development that's the an idea I'm not saying it's there it's just an idea uh so diversity and we've discussed this quite a bit today um insight into potentially gendered hunches I mean I'm super interested in that I might look this plays out when you look at the stats and things like stopping Vehicles it is different it's different from a agenda basis there's something going on no one's looking at it I think this this needs looking at um oper operationalization of hes and policing so um when and I'm currently planning a study that looks at this now so using both qualitative and quantitative methods investigating the factors that contribute to the activation and follow through of hunches but to do that you have to be aware that a hunch is happening and that's a that's the one of the things that we're we're trying to get our heads around like how do we how do we make officers aware enough that intuition is kicking in and then when that intuition is kicking in how do we help them describe it and also help describe their decision making off the back of it because I think again if we go back to that pragmatism side I think the hunch kicks in the police officer makes the decision

but there's very little rational consideration of the connection between the hunch and the action so that's something that I think we can look at um compar comparative analysis of policy models and I'll give you a really quick and dirty example of this so in the US obviously um it's a coercive policing model in the UK it's a consensual policing model both of those will change the way that hunches are generated they will um so have you heard of the touching the tail light on on police vehicle so in the US when um a cop touches uh stops a car as the police officer walks towards the car to speak to the driver they touch the tail light now there's lots of reasons for this the first is that you leave your fingerprint and DNA on the tail line okay but if you think about what that's doing in the officer's brain bra the immediate consideration when he gets out of that car is I could die and I need to leave evidence on this vehicle in front of me now I have stopped thousands of vehicles in my police career and I have never once thought that and because I've never once thought it I can hypothesize that the generation of hunches will be way different in terms of sensitivity to an American police officer's generation of hes not least because the threat is so mitigated in the UK the chances of me coming up against a firearm is almost nil I have never seen a gun I've never fired one until about six months ago so like 18 years of service and I've never carried and I've never seen a gun in a real life policing environment that's going to change the way that I police and I do think that there'll be a different operationalization of hunches as a result and that's something that would be a fascinating study and then um this one here this impact of technological support how does the use of Technology um change hunch development so for instance and this is going to sound really weird but let's just say a police officer is trained to record when a hunch is received and it could be as simple as a tap on the wrist for a wearable or something similar and uh that could then help us correlate the types of incidents that police officers are at when they receive hunches they don't have to explain them it's just like what where are you most likely to get them where does your intuition manifest you know and then we could look backwards at how it manifested and what it did and then the last one which is the public perception and Trust um possible to investigate how public perception and Trust in law enforcement are influenced by officers Reliance on hunches so I'll give you a quick example of this if you rely on hunches to do stop searches and that means that you do more stop searches that may affect Community Trust within that uh that particular community where you're conducting the stop searches that then becomes reciprocal so you rely on intuition to do stop searches rather than rational thought that leads to a rise in stop searches leads to a rise in arrests and also probably a rise in much more like um friction based interactions with members of the public long term that's

going to lead to a breakdown in in relationships and possibly um issues with legitimacy so how the public perceive um uh the legitimacy of police actions so that's just a an example of where we could go on the big scale this last one's a very macro like kind of level study and some of these up here are much more personally based and Daniel I think that's about it thank you thanks for over-viewing it I have some questions and there's some great questions in the live chat so okay that sounds good wow many places to jump in there's the fundamental tension with the possibly arbitrary or hard to explain yet intuitive and honed nature of the hunch and about how the hunch has such a rich before during and after there's kind of there must be something like hearing the code called on the radio yeah and that setting a stage like I'm going to a comedy I'm going to a horror movie or it's this code and that's all that is being passed through the blanket is just what code it is and that's equivalent to like saying what is your prior what do you prepare yourself for given that it's just this number or that number and what are the costs and the sensitivities to like false positives and false negatives so there's a lot of stages to investigate like looking forward to the Hun then looking back on it where certain kinds of rhetoric or certain kinds of thinking are privileged or supported by legal proceedings and saying something like well I just didn't think it was that way in the moment could be the most honest statement yet it might be seen as like almost malpractice okay awesome questions from live chat yes exactly right okay all right here we go I'll just read them um basically in the order they in all right so Susan asks does hunch lead to epistemic foraging or pragmatic foraging so let's just unpack that in the expected free energy decision making functional different policies are evaluated in terms of their epistemic value like expected Information Gain and pragmatic value which is aligning observations with your preferences so how does that raising of the hunch then get pushed one way or another into seeking more information epistemically or directly moving to take pragmatic action so I think that was a a very nice way of summing up a lot of the discussion that where I AI about the lean towards pragmatism the desire for control for a police officer when they first get there is like it's almost Primal and I know that's a big word to use um but they will I think tend to lean incredibly strongly into the pragmatic and then move into the epistemic so and if you think about this this mirrors the court process like if I if I turn up to an incident and there's threat within the incident the first thing that I want to do is and I know the word is awful because of the connotations it brings but uh neutralize the threat I want the threat gone because while the threat is there uh my intuition is going nuts I mean it overrides everything if I'm if I'm so I'll give you a very quick a practical example if I'm in an incident where I believe that the male party in domestic violence

is going to assault me or assault the victim while I'm there right the my brain does not want to allow any epistemic foraging none there's a safety override and the safety override is essentially make sure that this male is not going to assault you or assault the other person before we start asking questions and that safety override is often the thing that people will look at and say and this is what I meant when I talked about the difference between jurisdictions so I would say that the safety override in the US is much more powerful than it is in the UK because of the threat that's there because of the availability of firearms so you're probably going to find where there's more Firearms the link towards um uh pragmatic verigin is way higher way way higher and then once that's done neutralized whatever word you want to use then that's when the the longer the longer form questions and the much more like calm information gathering takes place so really good question there's definitely a lean uh towards pragmatic over epistemic but epistemic will happen and it just may come after the pragmatic so there could be a like kind of hierarchy uh a hierarchy if you like um performances because of the situation that presents does that make sense yeah like to give a kind of a non policing example it's like you dropped an object on the floor if it was very surely just soft carpet you might feel around blindly and epistemic foraging would have like a low risk but if you strongly believed that there was like other sharp objects on the floor you would be way more skeptical with just kind of reaching into dangerous locations that is Daniel can you answer my questions because that was way better than I just did okay next one we'll each give a shot we'll each give a shot um Susan asks how are police trained to avoid group think I.E not just follow the ones with improper force over Mor motivations So you you're talking about individual cognition in that moment but yet in that diad encounter there was like a collective intelligence component and a difference in hunches so how does the group training and operating come into play uh yeah this is really interesting because there's nothing about the way that your brain behaves under pressure in any of the police training nothing so and I suppose and this is a massive criticism and I and I used to talk about this a lot because I was involved in the design of What's called the pqf in the UK which is the education qualification framework um and it was a very very controversial project for me to be involved in but it was about raising the initial level of Education to degree level bachelor's level for police officers and one of the major drives is obviously to blunt to some extent the lean towards pragmatism and offer some prayers for the making of police decisions that are better than those based just in practice but this is a move away from what you would call like a kind of Master Apprentice model which is the uh and its experience-based epistemic model now of police learning into something

that's much more a priori which is um your more traditional educative professionalized model so are they taught to ignore group thing I'd probably say well there was no discussion of group thing during my training but I have to think back I mean my training was 23 24 years ago and I'm sure that things have progressed will it be good informed solidly evident stuff probably not and um I think in some cases I think there may be some really good academics who will be out there working with particular forces and they will be giving this sort of training but I think as a more General approach I don't think it's there and I think it's a massive Gap so really good question all right yeah very interesting it's um it's a team sport team art SL science and so it has to be at least considered how do officers share their hunches do they just raise their hand and say I have a hunch but that might be throwing a wrench into things in the moment um so H how do the right uh communication surface and then you have the the radio and you have the local communication you have these different Avenues of message passing but not all messages lead to resolution of the situation So Daniel that's a really little comment um we just submitted a paper which is on uh the use of decision making models in detective world and we were talking about this like I was only on a murder inquiry for a year of my service so um but I saw the way that the murder inquiries worked and it is way more rational and they invite hunches so like your senior investigating officers will say look have any of you got any feelings about this please try and explain them you know how or the or the other one where you just run an interview with one of the suspects and one of the first questions that would come out isn't what did the suspect say it's what do you think how did it make you feel you know so there's a a a real space within the detective world for the discussion and sharing of hunches now if I pick that same um uh instance up and you drop it into uniform because it's so temporarily enforced that you won't get Chance a lot of the time to share your hunches like so if I have a bad feeling about somebody at an incident and this is that lean towards pragmatism again I will want to make sure that I'm safe before I go anywhere I won't share with the officer next to me I have a bad feeling about that person I want to take pragmatic action to go and try and deal with that bad feeling so again this leans back into that other question that we've just discussed I would say that so the sharing of hunches and the passing of messages is probably much more available within the long the long decision making processes within detective work and very poor within the incredibly short temporally influenced decision- making of uniform does that make sense yeah it it also makes me think about active inference and statistics are providing like a science of uncertainty and decision-making amidst uncertainty and yet in many different domains me ranging from

policing to law science there's kind of bespoke ways of dealing with uncertainties and different settings even within the given field like different cases are going to have different treatments of uncertainties and so there's now this dance or Fusion to bring together those different treatments of uncertainty um I'm really curious how you see active inference and extended cognition in light of even though I'm not very familiar with the details the growth of policing Technologies and surveillance Technologies and the different paths that a given officer or Department might have in front of them in terms of how they incorporate technology into their whole apparatus so there's definitely going to be a really really big shift here um so when I left so this was two and a half years ago there were discussions three years ago about things like the integration of active heads up displays okay so when I'm at an incident if I've got facial recognition and it's running in the background and it's passing me information that I don't have and didn't have um how will that affect my decision making and that is and I'm going to be really honest really scary uh because really scary and really beneficial so I'll give you a quick example I turn up at an incident again I'm I'm going to keep using the domestic incident because it's simple and we and we as police officers go to them all the time okay so it's a and I keep calling myself a police officer I'm not a police officer anymore I'm full-time academic but you know long time so anyway um if I turn up at an incident and I have set of glasses in my body cam on my body Cam's running at all times so I know that there's some element of surveillance of my behavior and I know that whenever you look back at this incident my decision making is going to be um scrutinized okay so when it go when it goes into the court system my behavior will definitely be on show and I know that as a result obviously I'm I have to behave within the rules and the bounds and all that sort of thing now there's a propensity and I'm sure you know this for police officers to do what's called informal working practices where they essentially step outside the strict rules and laws and they try and take pragmatic decisions so um uh an example might be if I stop somebody speeding I might want to give them a warning I might not want to give them a ticket but speeding is technically a strict liability offense so I should have to give them a ticket and instead I operate and I use discretion I I decide not to do it I give words of advice and that person drives off okay with things like your heads up display if I'm walking towards that car and I'm wearing glasses that has given me full history full driving history including infraction speeding tickets Etc just by looking at the number plate and then when I get up to that driver's side window and the window comes down and my facial recognition tells me that this person has got previous for x y and Zed ass Soul Etc is that going to change my decision- making 100% because what what

has the police officer got to deal with now they they have what's in front of them and then they do that epistemic foraging the extended cognition that would come from Lifetime information during decision making is crazy um and I'll tell you right now we'll be doing it before we understand whether it is a good thing to do or not because that's what that's what police do they will jump in probably in fact I'd be surprised if people weren't testing this stuff now um uh and the other thing I think I mean I'm just going I went straight to wearable stuff there but there's other stuff that might help so if I'm thinking from an interceptive perspective there's a really good study um called emotional survival for law enforcement officers and it was a psych it was a police psychologist that wrote it and he put heart rate monitors on police officers and he showed that when you go into What's called the cold one incident in the UK which is an eer mer incident the blue lights are on you know like you were talking about the Cod the passing of the information and the the bodily reactions to it if I'm going to an emergency incident when I first start you're incredibly aware intercep ofely of what's going on so you can feel your heart you you your um breath quickens your uh periphery Narrows all of the normal stress um uh uh reactions kick in okay because you know that when you land there there's probably going to be conflict of some sort and you're going to have to deal with it now what the study showed was that very very experienced police officers so we're talking like 15 years 20 years plus when they were going to emergency incidents they would say that those bodily reactions were not happening but the um monitors would say that they were and this is super interesting because it may mean you you could say that the police officer has got more control over their interception or you could say the exact opposite so they have dulled their interception over time uh through exposure continued exposure and we don't know about the effects on hunches or the effects on General decision making in those different lengths of service if you like now if we go back to the extended stuff imagine having a wearable that tells you this lifetime that would be super interesting like it I mean again I'm like you know maybe 5 years in the future here but we're not a Million Miles Away there are well-being variables that are ongoing now I don't think they're being used for this purpose could they change decision making definitely will they and in good and bad ways hey Daniel this just needs a lot more research this bit this bit that you're discussing loads more research it just needs some proper proper good looking at uh because it could be immensely beneficial I can just see a million ways where it could also be quite damaging wow yeah the the awareness of the hunch and then there's one setting where the hunch is not aroused which may be like a burnout situation yeah exactly and then yet also on the other side of let's just say like a stage performer

they still might get the physiological arousal but they actually expect it and so they don't attend to it because this is the whole thing with we're not paying attention to what we expect yes and so they have a deeper experience layer that contextualizes the hunch and yet that might obscure that hunch from being communicated like to a younger colleague yeah and yet communicating where and how the experienced Professor or lawyer or officer feels in a situation is that ineffable experience transfer and then I I totally see these different paths with the wearables One path where the um boots on the ground are just reduced to an actuator in a decision-making system that they're not even a part of because it tells them here's what you want to do to this car don't epistemically forage just fulfill what the order is exactly and then there's the epistemic engagement mode and we can imagine like the family doctor asking somebody questions rather than like looking at their digital health record and there's kind of um there's a charming and an open-endedness to that and yet in high reliability situations people are going to be put into which lane are we going to take this doctor's office are we going to use digital health records or are we going to have slower potentially noisier encounter just because people self-report that they like that more than the other way yes yeah and just as you got me thinking there there's some really good studies that look at things like um uh cancer diagnosis and so you know somebody comes in with a bad ankle and a doctor says yeah you need to go for this test and this test and this test and they're unable to rationalize that it's the same thing but there's a much more so if you look at a doctor's purpose there they're still going to be pragmatically B they're going to be trying to solve whatever problem is is facing them um but their relevance realization has got so skilled over the years and this some of the stuff that you were just talking about there where there's lots of signals coming in that are just very very efficiently dealt with subconsciously and it may be about making that person aware of what that subconscious is doing in order to properly understand it and this is what I was talking about with the hunch Generation stuff because police officers don't often they don't interrogate the journey that led them to make the decision they make the decision and then they rationalize it backwards I think there's so much we could do here like there's so much potential because and again this is for improved decision making this isn't like um so we're looking at improved outcomes here we want to make policing better how do we make it better well I would say that increasing self-awareness about decision- making is absolutely fundamental you can't go anywhere this is just like perfect you know and the whole like the whole framework of active inference and um and the basing brain it just lends itself to this so well way better than the other theories that that I'd explored and some of which are in the

article like you know like the schema the scripts Etc um but I love ramstead work on nich construction in it that's something I think that I would like to look at a little bit more I think as as time goes on sorry my brain just went there towards the end I don't know why yeah I mean is there anything else than your question why have we got some more yeah yeah okay let's keep going with the questions um Jeff asks what overlap exists with military psychology zilon identifies active inference without directly referencing it in his recent book beyond the pale published by air University press I don't know the book but what do you think of the general question uh I don't but I have um so I work in a security and defense uh Academy so I'm surrounded by uh military Scholars uh that's a really good question and I will go and ask it for Jeff cool okay Dave asks how do hunches align with hallucinations in deep learning [Music] oh oh that's a really good question um and I have to be honest I mean I've read some stuff on hallucinations and I've listened to podcasts from machine learning Street talkers as well as some good good discussions around it I mean they're anomalies aren't they but and I suppose you're going to have a set of rules within the llm that still allows them to be displayed played in the same way that you're can have a set of rules within the body that allows the hes to come to the four I don't know is the answer to that um and I know that we don't properly understand hallucinations yet but I would I mean obviously we've got no consciousness in in in the the Deep learning machines point of contention but most people would say there's no consciousness there whereas the hunch the hunch is ability to rise through the Consciousness so the qualer of the hunch is the bit that makes it Salient and I think that that salience would be lacking in in any element of llm deep learning based work if that makes sense does that make sense Daniel yeah I I think uh another complimentary way that I thought of that question is the hallucinations let's just say in image output you say represent this and it makes a full resolution image that is non-f factually based and so people sometimes refer to it as hallucination so whereas the outputs of that are in terms of observations that are just factually ungrounded the hunch it's not like it's a it's not a visual phenomena it's not like there's a visual a new visual overlay or a sensor prediction it's actually something slower that is kind of like an upwelling of something that is Salient because as you pointed to it's slower it's interceptive it's affective it's not modifying like what we see smell hear in the way where when people talk about hallucinations as outputs of machine learning models usually might do that though Daniel that sounds really weird but I think it may to some extent do that so um and I'm just thinking about how that might be the case so if I'm at an incident I'll give I'll give you another very quick example so uh we went to um conduct a mandatory section in which is a

horrible incident and I hate them but it's essentially taking somebody into full-time mental health care he was a serious um schizophrenic with he was a danger to others um and we went to uh basically take him into custody so that he could go to the hospital um because he wasn't he wasn't refusing to take his medication or short for his appointment and it was caring the community Etc and he had attack somebody and I was in that room for probably one minute and I was telling the doctors to get out because I knew what was coming and if somebody had and this is what you're saying well it changes the way that you see well it does because it changes your relevance realization is that as the intuition kicks in and becomes stronger then your focus changes visually so you're going to be filtering out much more of the the side if you like and focusing in on what be what be considered the threat whatever the threat may be um and in this case I mean he was holding a knife down the side of a sofa and he was about to attack us and for some reason something about that incident again highly Salient feeling it then changed the way that I saw it changed the way that I interacted communicated changed the way that I dealt with my externals um uh and and I suppose you go call them variables so I wanted the other people out of the way because that would then give me greater uh Le that would give me less to think about when the conflict actually happened so I would probably say there's a little there's a parallel there I think that it can affect the senses and I imagine that no I quite like it it's an interesting thing to think about that's I saw in the paper you cited the the quantum reference frames and Chris fields's work so that just makes me think walking down a street one reference frame is just like oh there's people Milling about and then another reference frame whether this is paranoid or not is like their information to Circle me or somebody is shaking a little bit in their hand and it's like oh maybe they have essential tremor maybe they're nervous maybe they're about to do this and so that doesn't change the the photons hitting the retina that's right it changes our Framing and that is what is actionable and a more cognitively informed law and legal system could have a more agreed right grasp of this yes you're exactly right yeah yes very good I told you you're way better at this Daniel like you wrap up my words and it just comes out in a in a in a much simpler way very good let's we're we're cruising together um well where are you going to go in your continuing directions here like what are you going to do with this now that you have the paper okay so um there's a couple of things that I want to do I want to uh test um whether current decision- making models uh in policing that are used widely in many different countries um are useful for the processing and uh understanding of police hunches because I think that they are accountability models and not decision-making models so that's something that I want to look at but I

suppose the more uh relevant study will be researching the uh qualia of the hunch so um within different settings especially within uniform I'm very interested in it because of its contact with legitimacy as a whole like how police perceived I want to know how uniformed officers receive understand process and then act upon punches uh and uh the study that I have in mind is this is going to sound really weird um I want to do like a kind of personalized dictaphone like a narrative account so essentially when a police officer in any corner of the world makes a decision on the basis of a hunch that is then narrated into this kind of like um Repository and we use that repository as a qualitative um evidence-base to research how uh the hunch is process and then acted upon that's what I'm thinking so um I'm currently in the um process of getting funding and uh I'm hoping that I can secure it and if I can I'd like to run it as a this is going to sound really strange but a voluntary worldwide study so I don't care where you're a police officer um I want you to take part uh and we look at uh how you are receiving on a um a daily basis your minor hunches from I want to stop that car through to those like kind of heart hitting your stomach pyo erection High threat hunches that totally change the way that you see the incident in front of you so that's the that's the next step I think that makes me think of aroid which is a repository of alternate states of consciousness mostly based upon substances pyel yeah it's another but it's a it's a voluntary self report and that sounds very cool and voluntary I'm hoping I'm hoping that's the plan but um I'm hoping that once I get the funding that that will be a green light and obviously in the meantime I mean it's been so cool talking to you Daniel and I I know that the community is quite tight in around the active um uh INF Institute uh so if anybody has been listening or if anybody listens to this in the meantime if you've got any cool ideas or you want to ask me any questions look just email me I I I love to talk about this stuff in general and the discussion I know that we mentioned it right at the beginning but even the discussion where we talked about the law I've got I've got a theory that I've spoken to a professor about I'll just throw this out there just to finish off I think that people may call the police when the surprisal bounds are breached and they want the police to be an agent of return so essentially The Police become an agent of AI estasis to return to home estasis for that person now what that gives us and it's something that we don't have is a general theory of police contact and it allows us to see how different communities may call the police based on their different surprisal bounds so for instance in high chaos community is their sensitivity to surprise is going to be much higher than those that are in um sorry much lower than those that are in um uh very very structured gated communities for instance um and this this could lead us to some really interesting research but it's something that is very much

in the background of my brain and something that I'm I'm I'm thinking about a lot reading about a lot and hoping that maybe over time I can start to develop awesome thank you Gareth for um connecting these critical areas and bringing active in and it's just exciting to hear where it's going oh thank you Daniel thank you for your time and thank you to everyone who's actually managed to listen to me to drone along for this long it's it's been a pleasure to have you oh there's great comments in the live chat so thank you talk to you next time okay bye thank you very much bye bye see you for